

Chayan Roychoudhury

CONTACT INFORMATION

Modeling and Analysis of Atmospheric Composition Laboratory



EDUCATION

PhD, Atmospheric Science	The University of Arizona	2021 – 2025
Supervisor: Dr. Avelino F. Arellano Jr.		
MSc, Atmospheric Science	University of Calcutta	2017 – 2019
Thesis: Simulation of Hygroscopic Factors on Polar Aerosols over East Antarctica		
BSc, Physics (First Class Hons.)	University of Calcutta	2014 – 2017

WORK EXPERIENCE

Graduate Research Assistant	Jan 2021 - Present
Graduate Teaching Assistant	Jan 2023 - May 2024
ATMO 430 - Computational Methods in Atmospheric Sciences (Spring 2023)	
ATMO 569 - Air Pollution I : Gases (Fall Spring 2023)	
ATMO 545 - Introduction to Data Assimilation (Spring 2024)	
Department of Hydrology and Atmospheric Sciences, The University of Arizona	
Supervisor : Dr. Avelino F. Arellano Jr.	
Guest Research Worker	Aug 2019 - Jul 2020
Bose Institute, Environmental Sciences Section	
Supervisor : Dr. Sanat Kumar Das	

RESEARCH EXPERIENCE

Publications

1. C Roychoudhury, C He, R Kumar, AF Arellano Jr. 2025. **Diagnosing Aerosol-Meteorological Interactions on Snow within the Earth System: A Proof-of-Concept Study over High Mountain Asia.** *Earth System Dynamics*.
2. C Roychoudhury, C He, R Kumar, JM McKinnon, and AF Arellano. **On the relevance of aerosols to snow cover variability over High Mountain Asia.** 2022. *Geophysical Research Letters*, 49.
3. R Kumar, C He, C Roychoudhury, W Cheng, N Mizukami, and AF Arellano. **High Mountain Asia 12 km Modeled Estimates of Aerosol Transport, Chemistry, and Deposition Reanalysis, 2003-2019. (HMA2_MATCHA, Version 1).** 2024. [Data Set]. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. <https://doi.org/10.5067/CG40T8DJX2Z7>.
4. K Mottungan, C Roychoudhury, V Brocchi, B Gaubert, W Tang, MA Mirrezaei, JM McKinnon, Y Guo, DWT Griffith, DG Feist, I Morino, MK Sha, MK Dubey, M De Mazière, NM Deutscher, PO Wennberg, R Sussmann, R Kivi, TY Goo, VA Velazco, W Wang, and AF Arellano Jr. **Local and regional enhancements of CH₄, CO, and CO₂ inferred from TCCON column measurements.** 2024. *Atmospheric Measurement Techniques*, 17, 5861–5885.
5. M Greenslade, Y Guo, G Betito, MA Mirrezaei, C Roychoudhury, AF Arellano, A Sorooshian. **On ozone's weekly cycle for different seasons in Arizona.** 2024. *Atmospheric Measurement Techniques*, 17, 5861–5885.
6. MA Mirrezaei, AF Arellano, Y Guo, C Roychoudhury, A Sorooshian. **Ozone Production over Arid Regions: Insights into Meteorological and Chemical Drivers.** 2024. *Environmental Research Communications*, 6, 051009.
7. Y Guo, C Roychoudhury, MA Mirrezaei, R Kumar, A Sorooshian, and AF Arellano: **Investigating Ground-Level Ozone Pollution in Semi-Arid and Arid Regions of Arizona Using WRF-Chem v4.4 Modeling.** 2024. *Geoscientific Model Development*, 17, 4331–4353.
8. A Sorooshian, AF Arellano Jr, M Fraser, P Herckes, G Betito, E Betterton, R Braun, Y Guo, MA Mirrezaei, C Roychoudhury. **Ozone in the Desert Southwest of the United States: A Synthesis of Past Work and Steps Ahead.** 2024. *ACS ES&T Air*, 1(2), 62-79.
9. D Das, S Chiao, C Roychoudhury, F Khan, S Chaudhuri, S Mukherjee. **Tropical Cyclone Energy Variability in North Indian Ocean: Insights from ENSO.** 2023. *Climate*, 11, 232.

In progress

1. N Mizukami, S Minallah, C He, C Roychoudhury, W Cheng, R Kumar. **Hydrologic implications of aerosol deposition on snow in High Mountain Asia river basins.** *Environmental Research Letters*. (Submitted).

2. JM McKinnon, C Roychoudhury, MA Mirrezaei, B Gaubert, AF Arellano. **Use of Power Spectral Analysis and Dominant Time Scales to Investigate Seasonal and Long-Term Variability of Atmospheric Composition.** (*In preparation*).
3. JM McKinnon, C Roychoudhury, AF Arellano. **Spatio-temporal pattern analysis of MOPITT total column CO using varimax rotation and singular spectrum analysis.** [Atmospheric Measurement Techniques](#). (*In review*).
4. J Richter, E Joseph, M Arcodia, J Berner, J Demuth, P Falloon, G Romine, J Cohen, J Gonzalez-Cruz, M El Gharamti, B Hoppe, S Kumar, A Mariotti, D Mishra, K Pegion, Z Pu, K A Quagraine, K T Quagraine, C Roychoudhury, J Ryan, Z Stone, D Das, B Gaubert, S Kapnick, C Zarzycki. **Earth System Predictability Across Time Scales for a Resilient Society: A Research Community Perspective.** Bulletin of the American Meteorological Society. (*In review*).
5. C Roychoudhury, W Cheng, N Mizukami, C He, AF Arellano and R Kumar. **MATCHA, Model for Atmospheric Transport and Chemistry in Asia, a novel regional climate-chemical reanalysis. Part 1: System description and initial evaluation.** Earth System Science Data. (*In review*).
6. C Roychoudhury, C He, R Kumar, and AF Arellano. **Revisiting Black Carbon Emission Estimates in the Third Pole: A Hierarchical Bayesian Synthesis Inversion Perspective.** Environmental Research Communications. (*In review*).
7. H Bai, C Roychoudhury, C Strong. **Variability of summertime North American Dipole from large ensemble climate simulations.** (*In preparation*).
8. SK Das, C Roychoudhury, and A Taori. **Virga observed over East Antarctica: An alarming indication of global warming.** (*In preparation*).

TECHNICAL SKILLS

Programming Python, GrADS, IDL, MATLAB, QGIS/ArcGIS, PostgreSQL, L^AT_EX, Linux, HPC.
Modeling WRF-Chem, CESM/CESM-SCAM, MPAS.

HONOURS & AWARDS

- i) Recipient of John & Margaret Harshbarger Scholarship, University of Arizona (2024).
- ii) Recipient of Sol Resnick Scholarship, University of Arizona (2023).
- iii) Recipient of Galileo Circle Scholarship, University of Arizona (2022).
- iv) NSF NCAR Graduate Visitor Program (GVP) (2022).
- v) Rank 1 in MSc in Atmospheric Science (2019) and eligible for INSPIRE-Fellowship under DST, Government of India.

JOURNAL REVIEWER

- i) Geophysical Research Letters (Wiley).
- ii) Theoretical and Applied Climatology (Springer).
- iii) Environmental Monitoring and Assessment (Springer).
- iv) Journal of Applied Meteorology and Climatology (AMS).